Section 3.1 — Power and Radical Functions — Answer Key

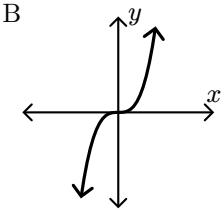
Math 103: College Algebra

Graphs of Power Functions

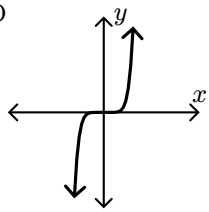
Match each function below to its graph.

A. $f(x) = x^2$ B. $g(x) = x^3$ C. $h(x) = x^6$ D. $k(x) = x^7$

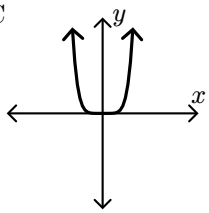
1. B



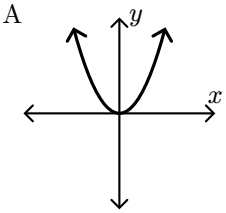
2. D



3. C



4. A



Domain and Range of Power Functions

Give the domain and range of each power function.

5. $\operatorname{Dom} f = (-\infty, \infty)$
   $\operatorname{Rng} f = (-\infty, \infty)$
6. $\operatorname{Dom} g = (-\infty, \infty)$
   $\operatorname{Rng} g = [6, \infty)$
7. $\operatorname{Dom} h = (-\infty, \infty)$
   $\operatorname{Rng} h = (-\infty, \infty)$
8. $\operatorname{Dom} k = (-\infty, \infty)$
   $\operatorname{Rng} k = (-\infty, 3]$
9. $\operatorname{Dom} p = (-\infty, \infty)$
   $\operatorname{Rng} p = [-4, \infty)$
10. $\operatorname{Dom} q = (-\infty, \infty)$
    $\operatorname{Rng} q = (-\infty, -8]$

Evaluating Roots

Fill in the blanks to evaluate each root by trial and error. (All of these are defined.)

11. $\sqrt{49} = \boxed{7}$ because $\boxed{7^2} = 49$
12. $\sqrt[3]{64} = \boxed{4}$ because $\boxed{4^3} = 64$
13. $\sqrt[6]{729} = \boxed{3}$ because $\boxed{3^6} = 729$
14. $\sqrt[5]{-32} = \boxed{-2}$ because $\boxed{(-2)^5} = -32$

Fill in the blanks to show the two consecutive integers each root is between by trial and error.

15. $\boxed{2} < \sqrt[5]{183} < \boxed{3}$ because $\boxed{32} < 183 < \boxed{243}$
16. $\boxed{8} < \sqrt{74} < \boxed{9}$ because $\boxed{64} < 74 < \boxed{81}$
17. $\boxed{5} < \sqrt[4]{731} < \boxed{6}$ because $\boxed{625} < 731 < \boxed{1296}$
18. $\boxed{-5} < \sqrt[3]{-99} < \boxed{-4}$ because $\boxed{-125} < -99 < \boxed{-64}$

Evaluating Roots on the Calculator

Evaluate each root using the calculator. Round your answers to three decimal places, if necessary. If the given root is undefined, write "undefined."

19. $\sqrt[6]{94} \approx 2.132$
20. $\sqrt{15} \approx 3.873$
21. $\sqrt{27} \approx 5.196$
22. $\sqrt[5]{-116} \approx -2.588$
23. $\sqrt[5]{11458} \approx 6.484$
24. $\sqrt[6]{-3323}$ is undefined
25. $\sqrt[7]{-66577} \approx -4.887$
26. $\sqrt[3]{237} \approx 6.188$
27. $\sqrt[4]{-554}$ is undefined
28. $\sqrt[3]{202} \approx 5.867$

Domain and Range of Radical Functions

Find the domain and range of each radical function.

29. $\operatorname{Dom} f = [\frac{1}{3}, \infty)$
    $\operatorname{Rng} f = [3, \infty)$
30. $\operatorname{Dom} g = (-\infty, -2]$
    $\operatorname{Rng} g = [-2, \infty)$
31. $\operatorname{Dom} h = [-6, \infty)$
    $\operatorname{Rng} h = (-\infty, -5]$
32. $\operatorname{Dom} k = (-\infty, \infty)$
    $\operatorname{Rng} k = (-\infty, \infty)$
33. $\operatorname{Dom} p = (-\infty, \infty)$
    $\operatorname{Rng} p = (-\infty, \infty)$
34. $\operatorname{Dom} q = (-\infty, 7]$
    $\operatorname{Rng} q = (-\infty, 1]$